Basics of natural language processing and unsupervised learning

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class 2

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Section 1

Intro

Overview

We are now familiar with algorithms for supervised machine learning — linear regression (for regression) and logistic regression (for classification). We know that it is important to split the data into a training set and a test set. To prevent overfitting, it may be important to use regularization for linear or logistic regression if the number of features is very large.

These models are trained by minimizing a suitable loss function that, roughly speaking, shows how the model's predictions deviate from the observed data.

Today we will continue with unsupervised learning.

Learning outcomes of today's class

1. Generate features from text data (document-term matrix).
2. Perform principal component analysis for dimension reduction and data visualization.
3. Explain K-means clustering and implement it manually.

Section 2

Natural Language Processing (NLP)

Text data

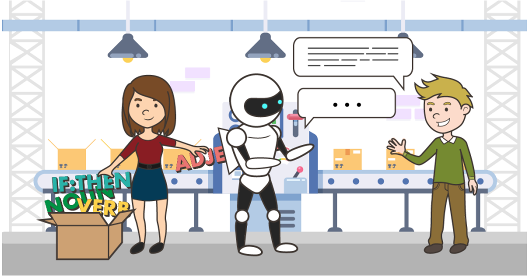
In natural language processing, we work with texts, e.g., Donald Trump's twitter:

text	retweet_count
@YouCanCallMeR @billmaher @jayleno Thanks.	2
@ChayalimBodedim Thank you.	1
@stkhlder3: @ApprenticeNBC @realDonaldTrump CAN NOT WAIT!!!!! fav is @BrandiGlanville !!!!!	10
Obama will be trying very hard at next debate- he doesn't want to lose the Boeing.	372
Congratulations Eric & Lara. Very proud and happy for the two of you! https://t.co/s0T3cTQc40	13712

NLP objectives

Sometimes we just want to make sense of the dataset (unsupervised learning).

Or we have some *downstream task* in mind — sentiment analysis (detecting mood in texts), document classification, question answering, named entity recognition, machine translation etc.



Activity: suggest some specific applications of natural language processing in finance.

- https://app.wooclap.com/ZETLIQ

Figure 1: WooClap activity

Text cleaning

Before working with texts, we usually need to do some cleaning.

Raw tweet:

text
Happy National Law Enforcement Appreciation Day! #LESM https://t.co/kNN3RetVvs

Cleaned text (removed all characters except for alphabet and numeric and changed everything to lower case):

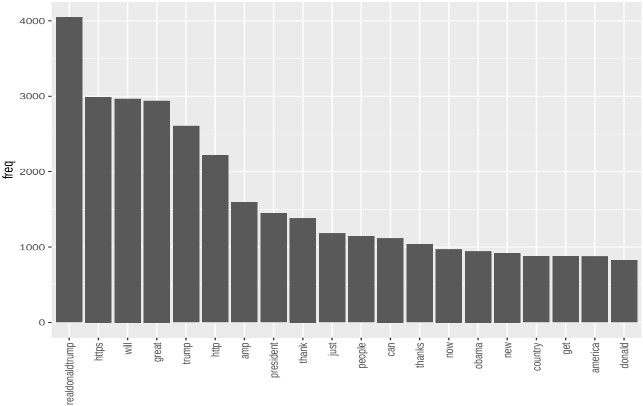
clean_text
happy national law enforcement appreciation day lesm https t co knn3retvvs

Often, we remove *stopwords* or do *stemming*. Depending on the task, one will choose a specific method for cleaning.

Text visualization

Word cloud is a nice way to visualize texts:

A simple bar chart is good too:



Document-term matrix

Our dataset is a collection C of documents, or a *corpus*. The set V of all the words or *terms* (i.e., depending on the task, punctuation may be counted as terms or removed) used in the corpus is the *vocabulary*.

The document-term matrix D is the matrix of size $|C| \times |V|$ with rows indexed by documents, columns by terms, and the entries count how many times a term occurs in a document.

$\dim(\mathrm{DTM}) = 20112\ 28337$

Usually we trim it down by removing non-frequent terms:

Removing terms that appear fewer than 80 times and stopwords: $\dim(\mathrm{DTM}) = 20112\ 461$

Then we can use columns of the DTM as features (predictors). There are some extra techniques to improve the DTM (e.g., TF-IDF, bigrams and trigrams). Anyway, the DTM will be a very sparse matrix and we will need some regularization to construct predictive models.

Here is a sample of the DTM:

	hillary	https	market	rally	stock
350	0	0	0	2	0
1460	2	1	0	0	0
1697	1	0	2	0	2
4511	0	4	0	0	0

And here is cleaned tweet 1697:

text
The Stock Market went up massively from the day after I won the Election all the way up to the day that I took office because of the enthusiasm for the fact that I was going to be President. That big Stock Market increase must be credited to me. If Hillary won - a Big Crash!

Notebook 3 (document-term matrix and regularization). Notebook 3:

- https://app.wooclap.com/ZETLIQ

Figure 2: WooClap activity

Section 3

Principal component analysis

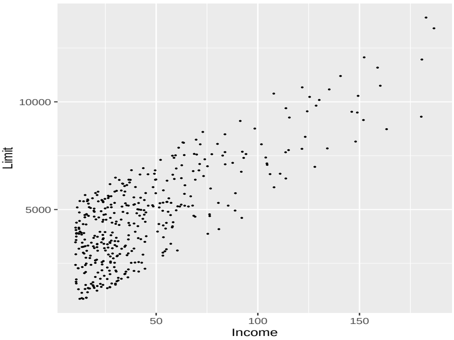
Motivation for PCA

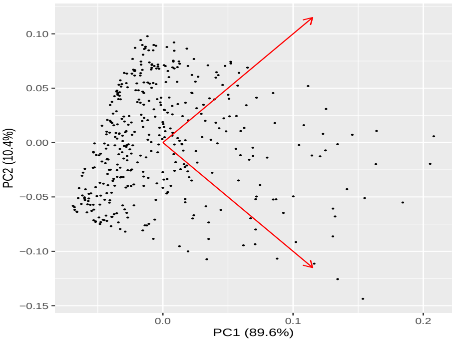
Principal component analysis is a technique for dimension reduction. Some reasons to do it:

- To visualize a multi-dimensional dataset, we often project it to a 2-dim plane, while trying not to lose too much information.
- Identifying hidden variables that affect observed variables.
- Interpreting coefficients in linear regression when there is multi-collinearity.

Principal component analysis replaces correlated variables with their linear combinations that are uncorrelated. It is similar to the Gram–Schmidt orthogonalization process in linear algebra.

Here is an example (credit card dataset from ISLR)





Before doing principal component analysis, we will shift the data so that all predictors have zero mean:

$$X_i \rightsquigarrow X_i - \bar{X}_i$$

We want to replace the original variables $X_1, \dots, X_p$ with their linear combinations

$$\begin{aligned} Z_1 &= \varphi_{11}X_1 + \varphi_{12}X_2 + \dots + \varphi_{1p}X_p \\ Z_2 &= \varphi_{21}X_1 + \varphi_{22}X_2 + \dots + \varphi_{2p}X_p \\ &\vdots \\ Z_p &= \varphi_{p1}X_1 + \varphi_{p2}X_2 + \dots + \varphi_{pp}X_p \end{aligned}$$

The ultimate goal is to produce uncorrelated new variables $Z_1, \dots, Z_p$.

Entries of the matrix $\Phi = (\varphi_{k,l})_{k,l=1}^{p}$ are called *loadings* and new variables $Z_1, \dots, Z_p$ *principal components*. Observations $z_i^1, z_i^2, \dots, z_i^n$ of a principal component are called *scores*.

The first principle component is required to have the largest sample variance, i.e., it solves the optimization problem

$$\frac{1}{n}\sum_{i=1}^{n}\left(\sum_{j=1}^{p}\varphi_{j1}x_j^i\right)^2 \rightsquigarrow \max \quad \text{subject to} \quad \sum_{j=1}^{p}\varphi_{j1}^2 = 1$$

Thus the first principal component represents direction in which the data vary the most.

In matrix form,

$$\varphi^T \mathbf{X}^T \mathbf{X} \varphi \rightsquigarrow \max \quad \varphi^T \varphi = 1$$

For example, let's look at our trimmed DTM for Trump's twitter. Then $\mathbf{X}$ is the de-meaned DTM, φ is the rotation matrix, and $\mathbf{Z} = \mathbf{X}\varphi$ is the matrix of new features.

dim(DTM) = 20112 461

dim(phi) = 461 461

dim(Z) = 20112 461

Sample of the rotation matrix φ (loadings):

	PC1	PC2	PC3	PC4	PC5
stock	0.00	0.00	0.00	0.00	0.00
market	0.00	0.00	0.00	0.00	0.00
hillary	-0.02	0.00	0.01	-0.02	0.00
rally	-0.01	0.00	0.00	0.00	0.00
https	-0.40	-0.63	-0.33	-0.05	-0.44

Correlations between some chosen old features:

	stock	market	hillary	rally	https
stock	1.00	0.64	0.00	0.02	-0.01
market	0.64	1.00	0.00	0.01	-0.02
hillary	0.00	0.00	1.00	0.00	0.01
rally	0.02	0.01	0.00	1.00	0.04
https	-0.01	-0.02	0.01	0.04	1.00

Correlations between new features:

	PC1	PC2	PC3	PC4	PC5
PC1	1	0	0	0	0
PC2	0	1	0	0	0
PC3	0	0	1	0	0
PC4	0	0	0	1	0
PC5	0	0	0	0	1

Biggest contribution to PC1:

	realdonaldtrump	will	https	trump
.	0.67	-0.42	-0.4	0.26

Biggest contribution to PC2:

	will	https	great	realdonaldtrump
.	0.68	-0.63	0.19	0.14

Biggest contribution to PC3:

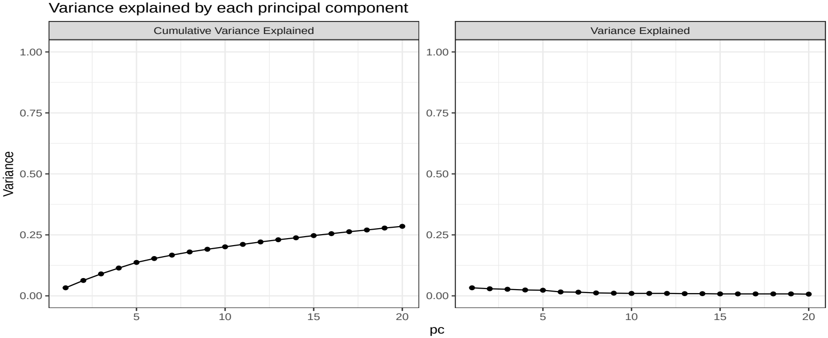
	realdonaldtrump	great	http	https
.	-0.53	-0.49	0.43	-0.33

Variance in the data explained by principal components:

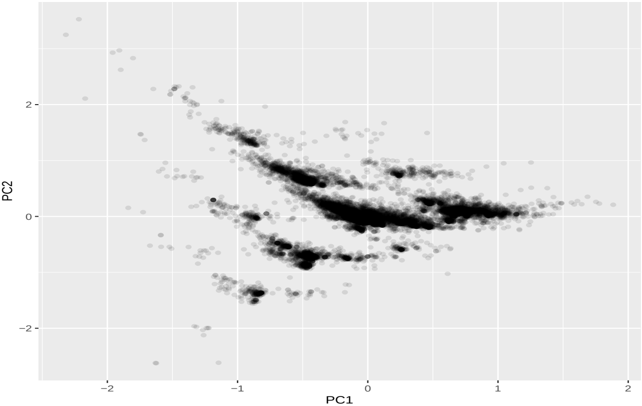
pc	variance	var_exp	cum_var_exp
1	0.204	0.033	0.033
2	0.180	0.029	0.063
3	0.165	0.027	0.090
4	0.146	0.024	0.114
5	0.141	0.023	0.137

To explain 70 % of variance in the data, we need 33 % of features.

Scree plot:


Variance explained by each principal component
Cumulative Variance Explained
Variance Explained
Variance
1.00
0.75
0.50
0.25
0.00
5
10
15
20
pc

Here is a plot of the entire twitter:



Quiz 1 (graded) - 30 minutes.

Section 4

Clustering

K-Means clustering

We want to partition our training data T into K clusters $C_1, C_2, \ldots, C_K$, i.e., find a representation as a disjoint union

$$T = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_K$$

such that observations within each cluster are similar to each other.

Let $W(C_k)$ be a dissimilarity measure of how observations within cluster k differ from each other.

There can be different ways to define $W(C_k)$. A common choice is the squared distance between data points x^i and $x^{i'}$, i.e.,

$$\|x^i - x^{i'}\|^2 = \sum_{j=1}^{p} (x_j^i - x_j^{i'})^2$$

Then we define $W(C_k)$ as

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \|x^i - x^{i'}\|^2$$

The optimization problem for K-means clustering is

$$\sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \|x^i - x^{i'}\|^2 \rightsquigarrow \min \quad \text{subject to} \quad T = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_K$$

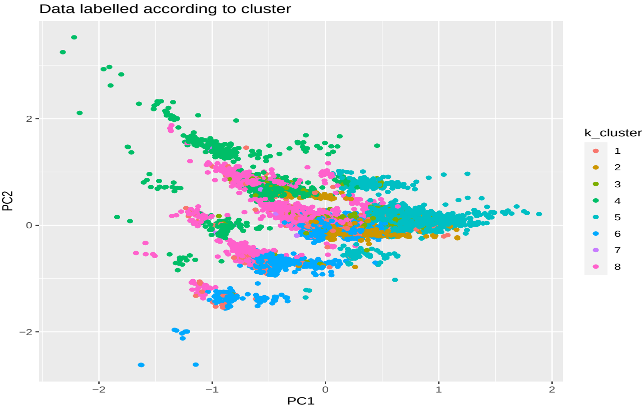
There is no solution in closed form and no efficient algorithm to find one, but there is a good algorithm to find a decent approximation to the optimum.

1. Randomly assign a number, from 1 to K, to each observation.
2. Iterate until the cluster assignments stop changing:

- For each of the K clusters, compute the cluster *centroid*, i.e, the mean of observations in that cluster.
- Assign each observation to the cluster whose centroid is closest.

The algorithm finds a local minimum rather than a global minimum. We can repeat it several times with different initial cluster assignments for a better result.

Here is what it looks like for the principal components of Trump's twitter:



Here is the frequency table:

	1	2	3	4	5	6	7	8
Freq	663	1810	1007	1800	3631	9219	39	1943

Some tweets in cluster 7:

text
How do you get Impeached when you have done NOTHING wrong (a perfect call) have created the best economy in the history of our Country rebuilt our Military fixed the V.A. (Choice!) cut Taxes & Regs protected your 2nd A created Jobs Jobs Jobs and soooo much more? Crazy!
Stock Market up big. New and Historic Record. Job jobs jobs!
....far safer and much less expensive. Many more cars will be produced under the new and uniform standard meaning significantly more JOBS JOBS JOBS! Automakers should seize this opportunity because without this alternative to California you will be out of business.

Section 5

Summary

Main takeaways

We are now familiar with document-term-matrix. It allows us to convert texts to numeric features.

Principal component analysis is a technique for dimension reduction, data visualization, and revealing hidden variables.

To do K-means clustering, we need to choose K, the number of clusters.

PCA:

- Parameters — the matrix of loadings
- Hyperparameters — the number of principal components to compute

K-Means clustering:

- Parameters — cluster centroids
- Hyperparameter — K
- Choice of dissimilarity measure

Concepts and ideas

1. Text to features: document-term matrix
2. Dimension reduction
3. PCA
4. Dissimilarity measure
5. Centroid
6. K-means clustering

At home

- Notebook 4 (Clustering, PCA, LDA)

Note that LDA is "Latent Dirichlet Allocation". Python codes for it are in Notebook 4 and you are encouraged to have a look at them, but the theory will only be explained in Lecture 3.